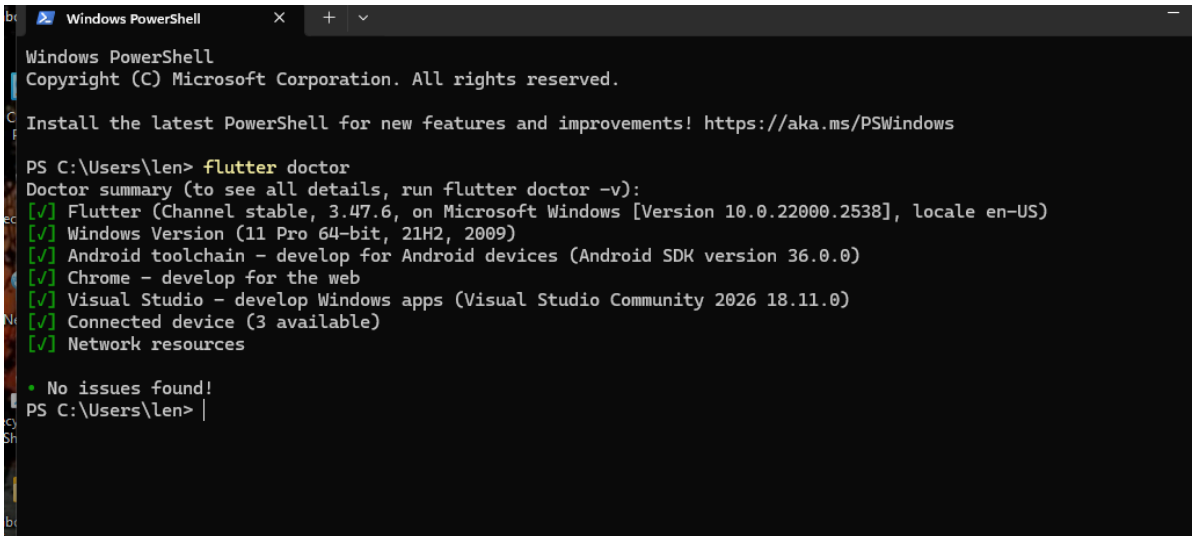
Run Hajir, 23/10/2026, ILO Cohort

TASK 1- Flutter & Dart Setup

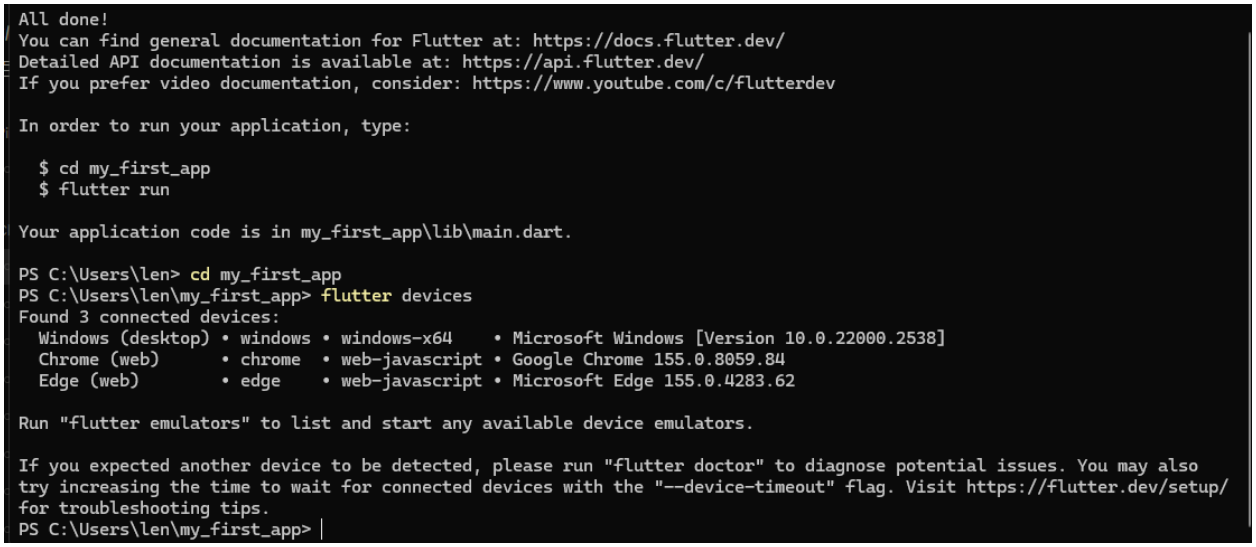
i) flutter doctor



2.ii) flutter create my_first_app



iii) flutter devices



iv) What is the difference between 'Hot Reload' and 'Hot Restart' in Flutter? Explain when a developer should use each.

- **Hot Reload** puts your new code into the running app in about a second and **keeps** what the app is remembering.
- **Hot Restart** starts the app again from zero and **resets** everything.

Task 2- MySQL Database Management

CREATE DATABASE school;

USE school;

CREATE TABLE students (

id INT AUTO_INCREMENT PRIMARY KEY,

name VARCHAR(100),

email VARCHAR(150) UNIQUE,

enrolled_on DATE

);

INSERT INTO students (name, email, enrolled_on) VALUES

('Amina Wanjiru', 'amina.wanjiru@example.com', '2026-09-01'),

('Brian Otieno', 'brian.otieno@example.com', '2026-09-02');

SELECT * FROM students;

```
mysql>
mysql> SELECT * FROM students;SELECT 1;
+----+---------------+--------------------------+-------------+
| id | name          | email                    | enrolled_on |
+----+---------------+--------------------------+-------------+
|  1 | Amina Wanjiru | amina.wanjiru@example.com | 2026-09-01  |
|  2 | Brian Otieno  | brian.otieno@example.com  | 2026-09-02  |
+----+---------------+--------------------------+-------------+
2 rows in set (0.026 sec)

+---+
| 1 |
+---+
| 1 |
+---+
1 row in set (0.001 sec)

mysql> SELECT * FROM students;
+----+---------------+--------------------------+-------------+
| id | name          | email                    | enrolled_on |
+----+---------------+--------------------------+-------------+
|  1 | Amina Wanjiru | amina.wanjiru@example.com | 2026-09-01  |
|  2 | Brian Otieno  | brian.otieno@example.com  | 2026-09-02  |
+----+---------------+--------------------------+-------------+
2 rows in set (0.003 sec)

mysql>
```

Security Reflection

Using the `root` account for an application backend is poor security practice because root has unlimited power over the whole MySQL server. It can read, change, or delete every database, and it can create or remove other users.

- **Bigger damage from attacks: if the application is attacked (for example through SQL injection) or its password leaks, the attacker gets full control of the entire server, not just one database.**
- **Bigger damage from mistakes: a bug in the application code could accidentally drop tables or databases that have nothing to do with the app.**
- **Breaks the principle of least privilege: an account should only have the permissions it needs to do its job, and nothing more.**
- **Harder to audit: when every application and administrator uses root, it is difficult to tell who did what.**

The safer approach is to create a dedicated user for the application that can only work inside the `school` database. The commands below are run once, logged in as root:

sql

```
CREATE USER 'school_app'@'localhost' IDENTIFIED BY
'Use_A_Strong_Password_1!';


GRANT SELECT, INSERT, UPDATE, DELETE ON school.* TO
'school_app'@'localhost';


SHOW GRANTS FOR 'school_app'@'localhost';
```

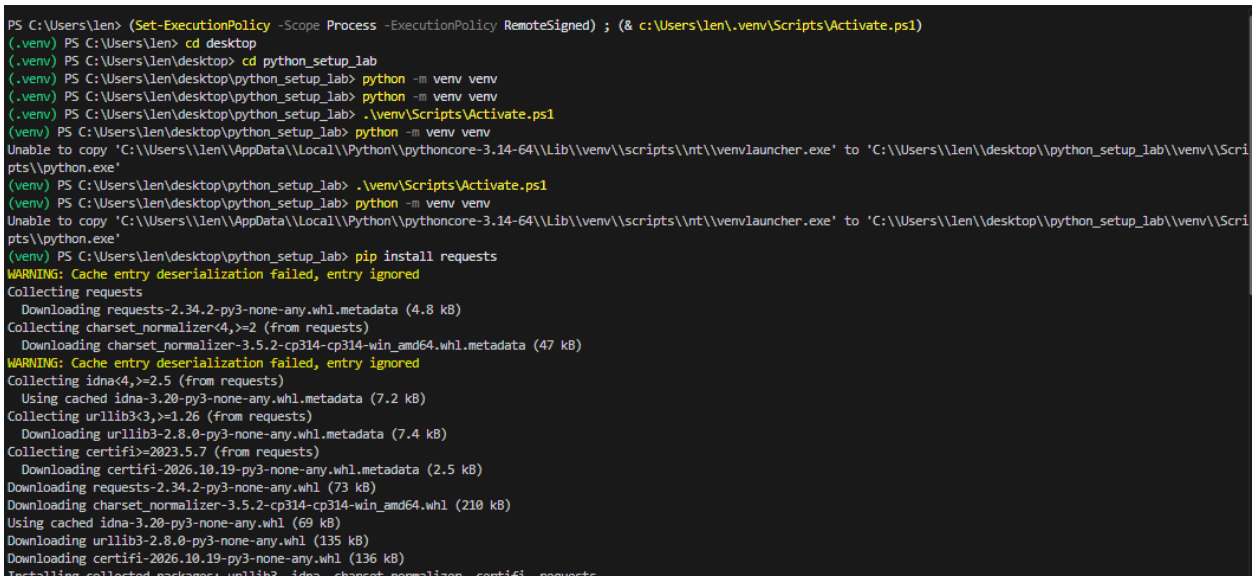
- **`CREATE USER` makes the new account. `'localhost'` means it can only connect from the same machine, and `IDENTIFIED BY` sets its password.**

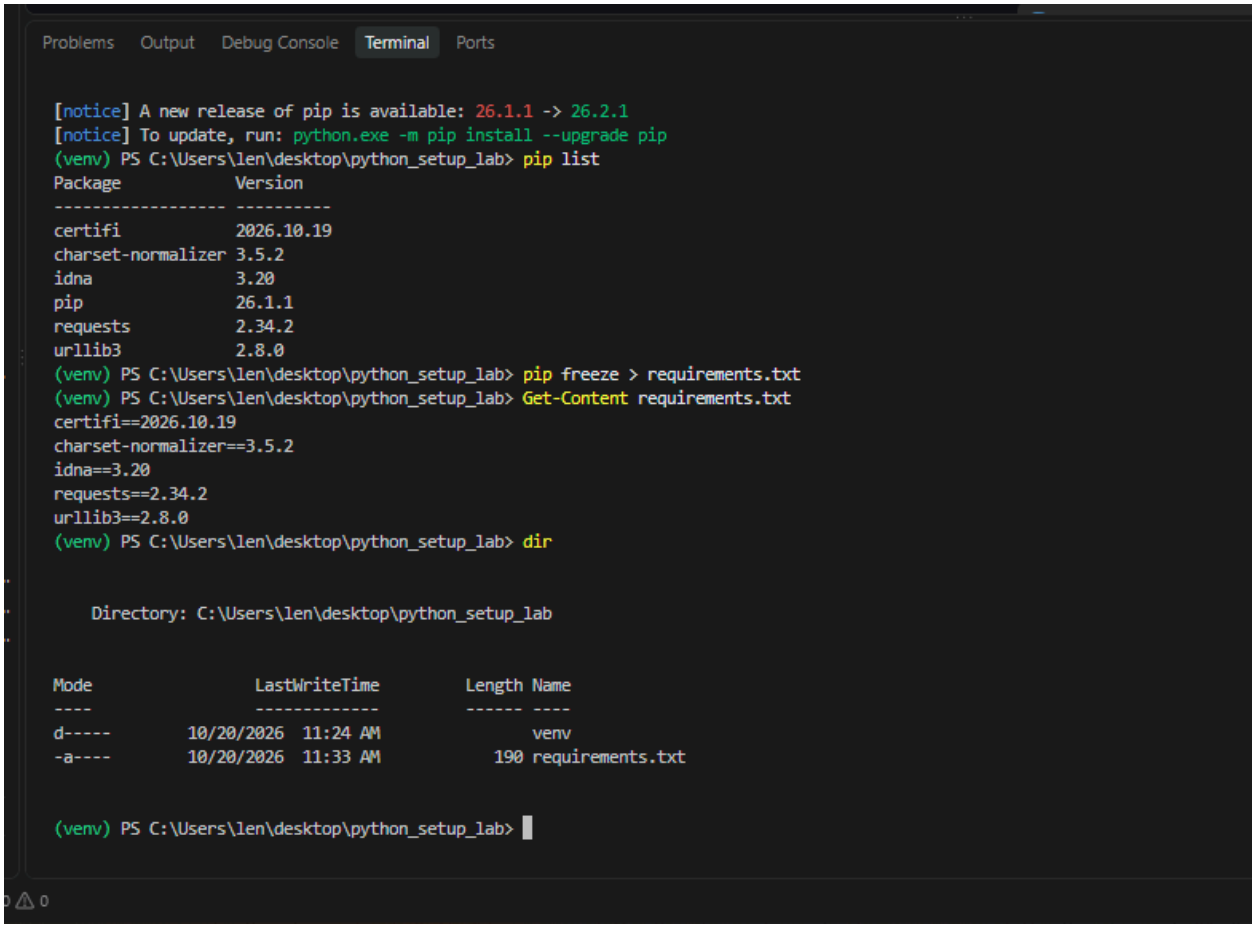
- **`GRANT SELECT, INSERT, UPDATE, DELETE ON school.*` allows the account to read and change rows in the `school` database only. It cannot create or drop tables, touch other databases, or manage users.**
- **`SHOW GRANTS` confirms exactly which permissions the account has.**

The application then connects as `school_app` instead of `root`, so a leak or bug can only affect the data in `school`.

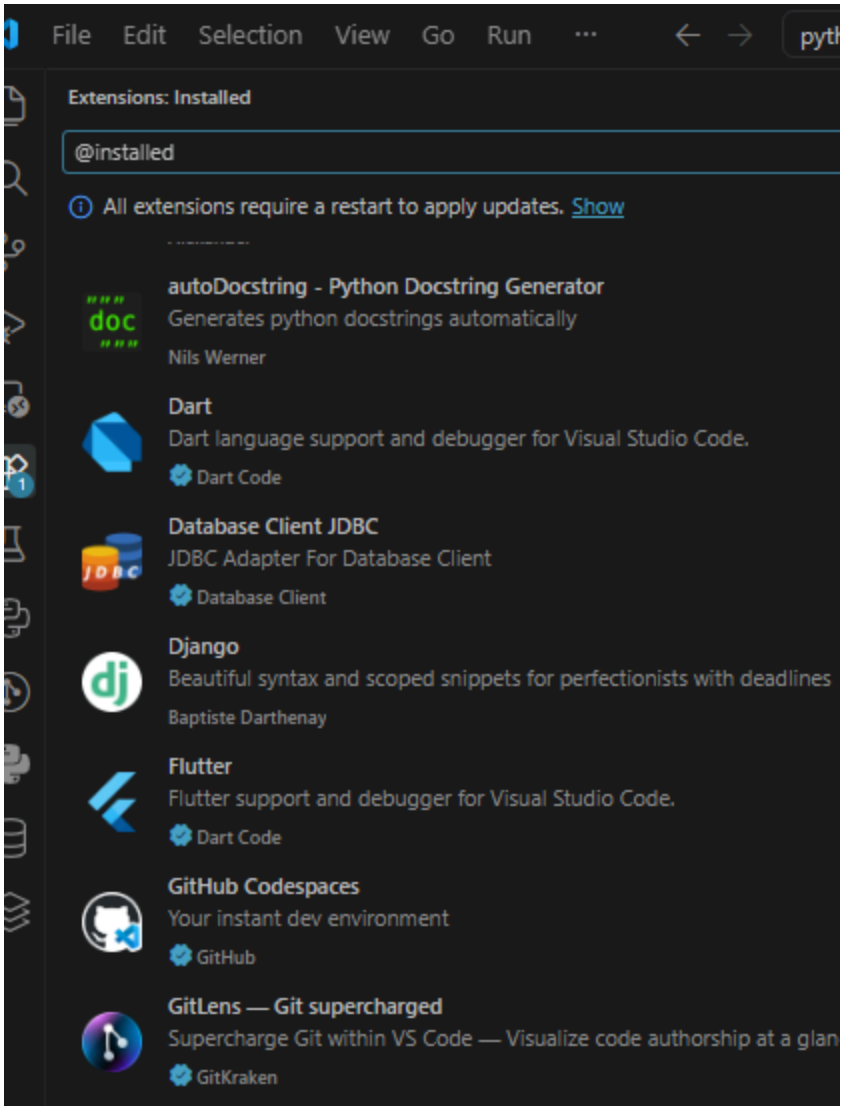
TASK 3- Python Virtual Environment

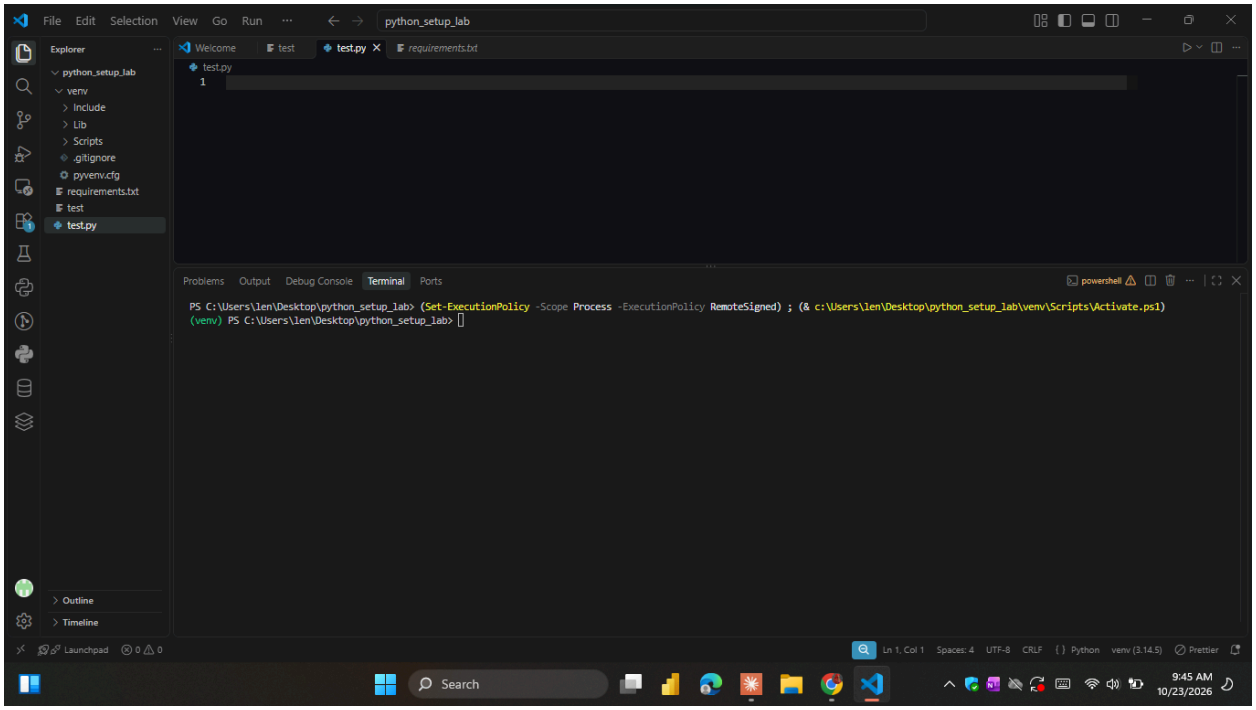
The terminal commands used for creation, activation and package installation.

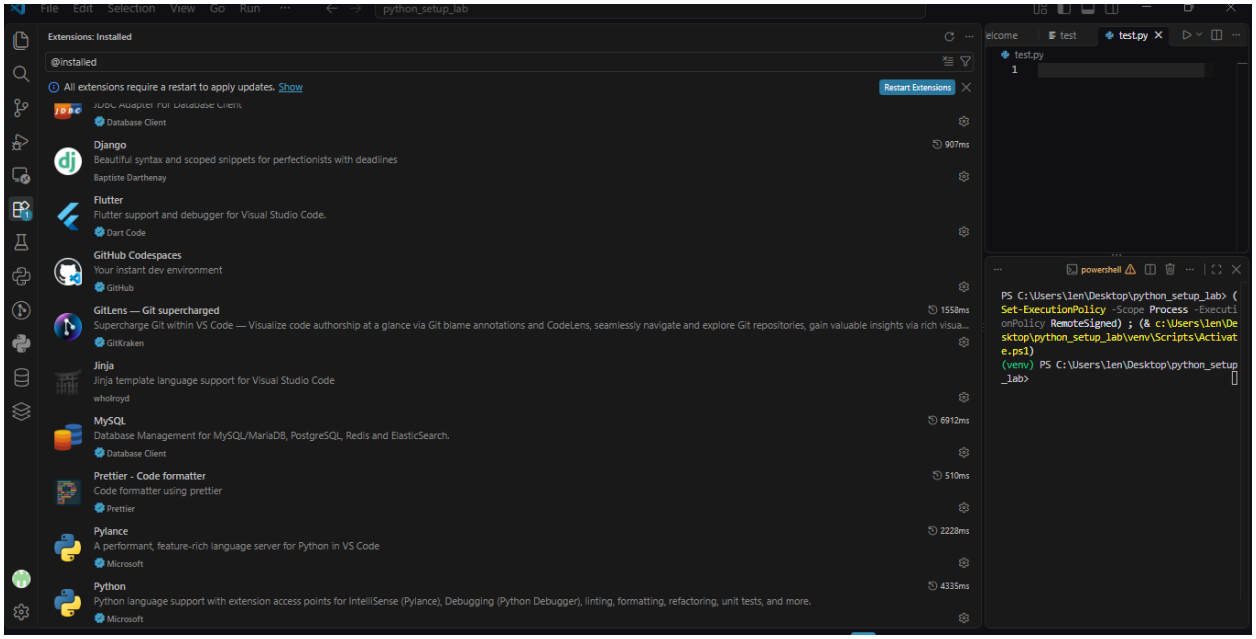




TASK 4- VS Code Workspace Screenshot


File Edit Selection View Go Run ···
Extensions: Installed
@installed
All extensions require a restart to apply updates. Show
autoDocstring - Python Docstring Generator
Generates python docstrings automatically
Nils Werner
Dart
Dart language support and debugger for Visual Studio Code.
Dart Code
Database Client JDBC
JDBC Adapter For Database Client
Database Client
Django
Beautiful syntax and scoped snippets for perfectionists with deadlines
Baptiste Darthenay
Flutter
Flutter support and debugger for Visual Studio Code.
Dart Code
GitHub Codespaces
Your instant dev environment
GitHub
GitLens — Git supercharged
Supercharge Git within VS Code — Visualize code authorship at a glan
GitKraken


File Edit Selection View Go Run
python_setup_lab
Explorer
python_setup_lab
venv
Include
Lib
Scripts
.gitignore
pyvenv.cfg
requirements.txt
test
test.py
Welcome
test
test.py
requirements.txt
1
Problems Output Debug Console Terminal Ports
powershell
PS C:\Users\len\Desktop\python_setup_lab> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ; (& c:\Users\len\Desktop\python_setup_lab\venv\Scripts\Activate.ps1)
(venv) PS C:\Users\len\Desktop\python_setup_lab>
Outline
Timeline
Launchpad
Ln 1, Col 1 Spaces: 4 UTF-8 CRLF Python venv (3.14.5) Prettier
Search
9:45 AM
10/23/2026


Extensions: Installed
@installed
All extensions require a restart to apply updates. Show
Restart Extensions
Database Client
Django
907ms
Beautiful syntax and scoped snippets for perfectionists with deadlines
Baptiste Darthenay
Flutter
Flutter support and debugger for Visual Studio Code.
Dart Code
GitHub Codespaces
Your instant dev environment
GitHub
GitLens — Git supercharged
1558ms
Supercharge Git within VS Code — Visualize code authorship at a glance via Git blame annotations and CodeLens, seamlessly navigate and explore Git repositories, gain valuable insights via rich visua...
GitKraken
Jinja
Jinja template language support for Visual Studio Code
wholroyd
MySQL
6912ms
Database Management for MySQL/MariaDB, PostgreSQL, Redis and ElasticSearch.
Database Client
Prettier - Code formatter
510ms
Code formatter using prettier
Prettier
Pylance
2228ms
A performant, feature-rich language server for Python in VS Code
Microsoft
Python
4335ms
Python language support with extension access points for IntelliSense (Pylance), Debugging (Python Debugger), linting, formatting, refactoring, unit tests, and more.
Microsoft
test.py
1
powershell
PS C:\Users\len\Desktop\python_setup_lab> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ; (& c:\Users\len\Desktop\python_setup_lab\venv\Scripts\Activate.ps1)
(venv) PS C:\Users\len\Desktop\python_setup_lab>